

This form documents the artifacts associated with the article (i.e., the data and code supporting the computational findings) and describes how to reproduce the findings.

Part 1: Data

- ☐ This paper does not involve analysis of external data (i.e., no data are used or the only data are generated by the authors via simulation in their code).
- ☒ I certify that the author(s) of the manuscript have legitimate access to and permission to use the data used in this manuscript.

Abstract

Our data has four components. The covariates, exposures, and adjacency matrix are provided along with the code to download the data. Health outcomes are not available due to data-sharing agreements, but we provided a simulated subset.

Availability

- ☒ Data **are** publicly available. Covariates, exposures, and shape files are available.
- ☒ Data **cannot be made** publicly available. Health outcomes are not publicly available.

If the data are publicly available, see the *Publicly available data* section. Otherwise, see the *Non-publicly available data* section, below.

Publicly available data

- ☒ Data are available online at: <https://www2.census.gov/geo/tiger/TIGER2023/ZCTA520/>
- ☒ Data are available as part of the paper's supplementary material.
- ☐ Data are publicly available by request, following the process described here:
- ☐ Data are or will be made available through some other mechanism, described here:

Non-publicly available data

The health outcomes are from Medicare data, which are not available to the public. Other parts of data, such as exposures and adjusting covariates, will be provided in the Github repo or directly downloaded from a R package. To allow for testing data analysis code, we provided simulated data for North Carolina. The R markdown file `data_analysis/data_analysis.Rmd` provides all steps to run models, collect results, and make plots.

Description

File format(s)

- ☒ CSV or other plain text.
- ☐ Software-specific binary format (.Rda, Python pickle, etc.): pkcle
- ☐ Standardized binary format (e.g., netCDF, HDF5, etc.):
- ☐ Other (please specify):

Data dictionary

The main files for data are ‘data_analysis/data_south.csv’ and ‘data_analysis/data_nc.csv’, with the latter being a subset of the former. The variable names should be self-describing.

- ☒ Provided by authors in the following file(s): Prim_SDMP.docx
- ☐ Data file(s) is(are) self-describing (e.g., netCDF files)
- ☐ Available at the following URL:

Additional Information (optional)

Part 2: Code

Abstract

Code is provided in two folders: `data_analysis` and `simulation`. For data analysis code, `data_analysis.Rmd` includes all steps to run models and see results, and `functions_analysis.R` includes all necessary functions. For simulation code, we provide the functions needed in `functions_sim.R`, simulation code (`simulation.R` and `simulation_nonlinear.R`), a script to send the simulation code to an HPC cluster (`submit.csh`), an Rmarkdown file to compile results and make plots (`sim_result.Rmd`), and a PDF of the Rmarkdown output (`sim_result.Rmd`).

Description

Code format(s)

- ☒ Script files
 - ☒ R
 - ☐ Python
 - ☐ Matlab
 - ☐ Other:
- ☐ Package
 - ☐ R
 - ☐ Python
 - ☐ MATLAB toolbox
 - ☐ Other:
- ☐ Reproducible report
 - ☐ R Markdown
 - ☐ Jupyter notebook
 - ☐ Other:
- ☐ Shell script
- ☐ Other (please specify):

Supporting software requirements

All code is in R. The R packages used are listed below.

Version of primary software used

- R version 4.2.2

Libraries and dependencies used by the code

- findSVI 0.2.0
- sf 1.0-14
- tidyverse 2.0.0
- spdep 1.3-1
- MCMCpack 1.6-3
- MASS 7.3-60
- splines2 0.5.1
- plyr 1.8.9
- LaplacesDemon 16.1.6
- doParallel 1.0.17
- fields 15.2
- spam 2.10-0

Supporting system/hardware requirements (optional)

Parallelization used

- ☐ No parallel code used
- ☒ Multi-core parallelization on a single machine/node
 - Number of cores used: 20 (for simulation only)
- ☐ Multi-machine/multi-node parallelization
 - Number of nodes and cores used:

License

- ☒ MIT License (default)
- ☐ BSD
- ☐ GPL v3.0
- ☐ Creative Commons
- ☐ Other: (please specify)

Additional information (optional)

Part 3: Reproducibility workflow

To reproduce data analysis, the most straightforward way is to run `data_analysis/data_analysis.Rmd`. All files needed for this script are provided.

Scope

The provided workflow reproduces:

- ☐ Any numbers provided in text in the paper
- ☒ The computational method(s) presented in the paper (i.e., code is provided that implements the method(s))
- ☐ All tables and figures in the paper
- ☒ Selected tables and figures in the paper, as explained and justified below: The provided workflow reproduce (a different version of) Tables 1 and 2 and Figures 4 and 5. These are the key results for data analysis. Note that the tables and figures will be different from those in the paper, since the outcomes are simulated and only for North Carolina.

Workflow

Location

The workflow is available:

The workflow is described in this current document.

- ☐ As part of the paper's supplementary material.
- ☐ In this Git repository:
- ☐ Other (please specify):

Format(s)

- ☒ Single master code file `data_analysis/data_analysis.Rmd` has all the code for running data analysis.
- ☐ Wrapper (shell) script(s)
- ☐ Self-contained R Markdown file, Jupyter notebook, or other literate programming approach
- ☐ Text file (e.g., a readme-style file) that documents workflow
- ☐ Makefile
- ☐ Other (more detail in *Instructions* below)

Instructions

All code for data analysis is provided in `data_analysis/data_analysis.Rmd` and all files needed are provided in the same folder.

Expected run-time

Approximate time needed to reproduce the analyses on a standard desktop machine:

- ☐ < 1 minute
- ☐ 1-10 minutes
- ☒ 10-60 minutes If a subset with simulated outcomes is run with 10,000 iterations, as set up in `data_analysis.Rmd`, less than an hour is needed. But the computation time for the full data analysis with 100,000 iterations would take more than 24 hours.
- ☐ 1-8 hours
- ☐ > 8 hours
- ☐ Not feasible to run on a desktop machine, as described here:

Additional information (optional)

Notes (optional)